

Domande (con risposte) dai Temi d'Esame di Robotics

MAIN PROBLEMS:

- a) Provide the derivation of the recursive Bayes Filter for Online SLAM
- b) Describe the Bayes filter algorithm for Online SLAM

17/07/2017

1.1 Question:

Let's consider the problem of obstacle avoidance, a.k.a., local path planning, and the algorithms to implement it. Provide a description of

- a) the aim of local path planning and what should be its frequency with respect to global path planning (explain the reason for that!)
- b) The functioning of the Vector Field Histogram approach (VFH) for obstacle avoidance, and its improvement Vector Field Histogram+ (VFH+)
- c) The Dynamic Window Approach (DWA)
- d) All these 3 algorithms use a navigation function, i.e., a function evaluating the value/cost of possible actions. Describe such navigation function and the parameters involved in it. Discuss briefly how it could be possible to set up such parameters in practice.

Answer:

- a) Local path planning's aim is to follow the planned path and to avoid unexpected obstacle that are not indicated in the map (e.g. a person walking). Local planning must be done more frequently than Global Path planning as it's useful to review the local environment and making short-time decision more often than decide and plan the full trajectory that the robot has to walk through to reach its goal.
- b) VFH uses a two Degrees-Of-Freedom local map of the environment and evaluates the angle to drive towards, so that all openings for the robot to pass are found: the one with lowest cost will be selected. VFH+ instead considers robots moving on arcs and not only on straight line and its thought for vehicle kinematics. Obstacles here are also enlarged so to account for all kinematically blocked trajectories.
- c) The Dynamic Window Approach algorithm is made up of four major steps:
 1. Gathering every circular trajectory (defined by a pair of angular and linear velocity $[w, v]$) that the robot can perform.
 2. Keep just the admissible trajectories, meaning that the robot is able to stop before the closest obstacle.

3. Choosing the trajectories that can be reached within a short amount of time, these trajectories are called reachable.
4. Choosing the solution with the lower cost based on the navigation function, discarding “illegal” trajectories such as those who collide with obstacles.

d) $G = a \cdot \text{target_direction} + b \cdot \text{wheel_orientation} + c \cdot \text{previous_direction}$. The three aspects taken into consideration are TARGET_DIRECTION: alignment of the robot path with the goal; WHEEL_ORIENTATION: the orientation that the wheels have to keep in order to continue the movement; PREVIOUS_DIRECTION: the direction that the robot had before the current time. a, b, c are the parameters that give the right way to the correspondent aspect and the best way to set up them in practice is in a heuristic way, meaning that they will need to be determined depending on the situation we are facing.

1.2 Question:

In the figures below you can appreciate the same vehicle but with two different kinematics. For each picture describe: the type of kinematic of the vehicle (we have seen both during classes!), their direct kinematic equations (in terms of the control variables), the advantages/disadvantages of each. (Picture: see on the pdf)

Answer:

The two types of kinematics are for the first one Ackermann Steering (bicycle approximation) and Differential Drive Kinematics. Their direct kinematics equations are the following:

DDK:

$$x(t) = 1/2 \int (V_R(t) + V_L(t)) \cdot \cos(\theta(t)) dt$$

$$y(t) = 1/2 \int (V_R(t) + V_L(t)) \cdot \sin(\theta(t)) dt$$

$$\theta(t) = 1/L \int (V_R(t) - V_L(t)) dt$$

V_R = velocity of the right wheel, V_L = velocity of the left wheel

ACKERMANN:

$$x(t) = 1/2 \int (V_F(t)) \cos(\theta(t)) dt$$

$$y(t) = 1/2 \int (V_F(t)) \sin(\theta(t)) dt$$

$$\theta(t) = 1/L \int (V_F(t)) dt$$

V_F = Velocity of the anterior wheel (the posterior is passive)

The key difference between these two types of kinematic is that Ackermann takes into consideration the steering angle of the anterior wheel when computing the frontal speed of it.

The main advantage of DDK is that it has simple equations, while the main disadvantage is that a robot built with that kinematics can't move laterally.

Neither an Ackermann vehicle can move laterally, but the main advantage it has is that the non-approximated model (a robot with four wheels with different orientations and different speed) is the most diffused kinematic on the planet. Furthermore, a robot built in that way can't perform in place rotation, while a DDK one can.

1.3 Question:

In ROS you have topics, services and actions, (a) what are they used for, (b) what are the characteristics of each of them in terms of usage, blocking/asynchronous behaviour?

Answer:

a) Topics: named buses over which nodes exchange messages. A node can subscribe (subscriber) to a topic in order to read the messages stored in it or it can publish (publisher) messages on a topic so that other nodes can read those messages.

Services: a providing ROS node offers a service under a string name, and a client calls the service by sending a request message and awaiting the reply. Services are defined using ".srv" files. Services are not meant for long execution time, they cannot be monitored nor cancelled.

Actions: similar to services but they are meant for long execution time because they can be monitored and cancelled. Actions are used when the requesting node cannot wait.

b)

1.4 Question:

Describe the Beam Sensor Model in terms of (a) its components and the rationale behind each of them, (b) its use in a particle filter used for Monte Carlo localization (to answer properly this point you need to first briefly describe particle filtering and Monte Carlo localization).

Answer:

a) The Beam Sensor Model is composed by four different types of probabilities that have to be mixed to make it work. First of all, we have a Gaussian density that represent our measurement if we were in an ideal world: the sensor measures the distance of the obstacle, but the value returned is subjected to error, modelled by a Gaussian. Secondly, we have an Exponential density that represents the possibility of sensing an unexpected object that isn't contained in the map we have in our possession (e.g. a person walking); it is exponential as the probability of this decreases with range, and is "cut" because a stranger object can't be sensed if it is more distant than the obstacle. Finally, we have two uniform probabilities, one that represents the failure of our measurement (the

obstacle isn't sensed and the sensor returns the max value it can measure) and another one the random errors (a sensor randomly produces entirely unexplained measurements). PUT HERE THE REPRESENTATION OF THEM.

b) Particle Filter is an application of Bayes Filter, that approximates the posterior belief of a robot position by a set of random samples called particles, each of them with a specific weight that represent the likelihood of it to be the right position of the robot. In Monte Carlo Localization (an application of Particle Filter) the starting weight of these particles is equal and it gets recursively updated while the robot moves and senses something; finally the particles will converge towards the actual robot position and the more the robot moves into the environment, the more precise is its knowledge about its position. *Here the Beam Sensor Model is used by the robot to sense the environment around him while he's moving (??? NOT SURE)*

01/07/2017

2.1 Question:

With reference to the synchronous drive kinematics, answer the followings: a) Describe a robot with an omnidirectional robot kinematics b) The synchronous drive kinematics is holonomic, what does this mean? c) Describe what are direct and inverse kinematics (in general) and what they are used for d) Assuming a synchronous drive robot starts in $[x_s, y_s, \theta_s] = [0, 0, 0]$ compute the inverse kinematics, i.e., the setpoints for linear and angular velocities, to let the robot reach the goal position $[x_g, y_g, \theta_g] = [\sqrt{3}, 1, \pi/2]$. Assume the maximum linear speed is 1m/s and angular speed 1rad/s.

Answer:

- a) A robot with an omnidirectional kinematics is designed with three Swedish Wheels each of them actuated with an independent motor. All of them point in different direction and it's possible to control x, y, theta.
- b) "Holonomic" means that the robot can move in any direction, meaning that it can exploit all 3 degrees of freedom of the space where it's in.
- c) Direct kinematics is the problem of finding a robot final position given the beginning one and the control variables, that are linear and angular velocities. Inverse kinematics is the opposite; given the starting and final position, it's needed to find the linear and angular velocity to reach the goal.
- d) Because of the fact that computing inverse kinematics is more difficult than doing the same for direct, it's useful to decompose the problem into three steps:
- Turn the robot's wheels so that they are parallel to the original and final position; we can do it giving it the maximum speed for $\pi/6$ seconds: the robot's final position will be set in $(\sqrt{3}, 1)$ meaning that the hypotenuse of the triangle is 2 and so the angle between the original state and the parallel to the wheels is 30 degrees.
 - Drive straight to the final position: we can do it giving it the maximum speed for 2 seconds.
 - Rotate again to achieve the desired orientation: following the first point description, it will be for another $\pi/3$ seconds.

2.2 Question:

Let's consider the Simultaneous Localization and Mapping problem; answer the following questions: a) How Simultaneous Localization and Mapping (SLAM) differs from Localization? b) How Simultaneous Localization and Mapping (SLAM) differs from Mapping? c) Describe what is the difference between online and offline SLAM d) Describe how scan matching works, and describe its limits e) Provide the derivation of the recursive Bayes Filter for online SLAM

Answer:

- a) SLAM differs from localisation because the map is unknown and has to be estimated along the way.
- b) SLAM differs from mapping since the poses of the robot are unknown too and have to be estimated along the way.

c) Both types of SLAM have to estimate the map. The difference is about the robot's position: SLAM involves the estimation of it only at current time, while offline SLAM has to find the entire path from the beginning to the current time.

d) Scan matching is a method useful to incrementally align two scans, to maps or a scan and a map without revising the past. Basically, it maximizes the likelihood of the current pose and map relatively to the previous pose and map

$x_t^* = \operatorname{argmax}\{p(z_t | x_t, m_{t-1}) * p(x_t | u_{t-1}, x_{t-1}^*)\}$. After updating current position in this way, we'll have to update the map using the new updated data so that in $t+1$ we can use it and so on...

Its main limits are the fact that lasers used to perform it are subjected to errors (such as noise) and that moving an object between two scans could mean that it won't be visible in the second or wasn't visible in the first (Not sure: found it on the internet).

e)

2.3 Question:

Describe briefly (a) what are ROS and (b) Gazebo, (c) what is their relationship and (d) how they can be used together. *DO WE HAVE TO DO IT?*

Answer:

a) ROS is a collection of software frameworks for robot software development.

b) Gazebo is an open-source 3D robotics simulator

c)

d) They can be used to simulate both the robot behaviour in a given environment (gazebo) and the algorithms that make the robot move and map the environment (ros).

2.4 Question:

Describe (a) how an electric DC motor works, (b) what gears coupled with a DC motor are used for, (c) what rotatory encoders coupled with DC motors are used for.

Answer: *(too short: missing something?)*

a) An electric DC Motor converts electricity into mechanical energy. A current passes through loops of wires mounted on a rotating shaft; these wires create a magnetic field which reacts against the one created by magnets that are placed around them. The interaction between them makes the armature of the motor turn and so mechanical energy is formed.

b) Gears are really important because usually a motor turns with 3-9000 RPM which is too much. By combining gears with different radius, we can control the Force and the Torque, so that the motor will turn slower and will last more.

c) Rotary encoders are used to convert the position or the rotary motion of a motor into an electronic pulse.

26/09/2016

3.1 Question:

With reference to a differential drive robot having a wheel base width of 50cm, answer the followings: a) Describe what are direct and inverse kinematics and what they are used for b) Provide the direct kinematics formulas for a differential drive robot c) Assuming the robot starts in $[x_s, y_s, \theta_s] = [0, 0, 0]$ compute the inverse kinematics, i.e., the wheel speed setpoints, to let the robot reach the goal position $[x_g, y_g, \theta_g] = [\sqrt{3}, 1, \pi/2]$. Assume the maximum speed for each wheel is 1m/s.

Answer:

a) See 2.1, c).

b) See 1.2

c) Similar to 2.1 d), with a key difference: $w(\omega) = (V_R - V_L)/2$. The rest of the exercise gets resolved in the same way, decomposing the movement.

3.2 Question:

A robot is filled up with sensors, some of them are proprioceptive others are exteroceptive: a) What is the difference between proprioceptive and exteroceptive sensors? b) Provide one example of sensor for each category c) What are encoders used for? d) Are encoders proprioceptive or exteroceptive sensors? e) Describe the difference between linear and rotary encoders f) Describe the difference between incremental and absolute encoders g) How does the quadrature technique work? h) How does absolute rotary encoders work?

Answer:

a) Proprioceptive sensors are used by the robot to sense its internal state; exteroceptive instead are used to sense the environment around it.

b) Proprioceptive: gyroscopes, accelerometers, magnetometers, encoders...

Exteroceptive: sonar, time-of-flight camera, time-of-flight telemeter...

c) See 2.4 c)

d) Encoders are proprioceptive.

e) A linear encoder consists of a long linear read track, together with a compact read head and is used to convert the position of the robot; the rotary one instead has a rounded shape and it converts the angular position of the robot.

f) An incremental rotary encoder is disk with two traces (or sensors) where transparent and opaque zones alternate, while the absolute rotary one is formed by transparent and opaque areas but on concentric rings (for an N-bit word you'll need N rings).

g) The quadrature technique is used in incremental rotary encoders to increase their resolution. It consists in shifting the two signals by $\frac{1}{4}$ step and the values of two signals indicate how the encoder is

turning. In particular, if 11 is followed by 10 means that it's moving counter clockwise, while if it's followed by 01 means that it's moving clockwise.

h) ??

3.3 Question:

Describe briefly (a) what are ROS and (b) Gazebo, (c) what is their relationship and (d) how they can be used together.

Answer:

See 2.3.

3.4 Question:

Describe (a) what is scan matching, (b) how it works, and (c) what it is used for.

Answer:

See 2.2, d): too short? What could be added?

05/09/2016

4.1 Question:

Let's consider Simultaneous Localization and Mapping, in particular: a) Describe the Simultaneous Localization and Mapping problem b) Describe the difference between online SLAM and full SLAM c) Describe what a particle filter is and how it can be used for SLAM or localization d) Describe the FastSLAM algorithm for full SLAM

Answer:

a) The SLAM problem arises when the robot doesn't have access to the map of the environment neither to its position and so to compute both of them while moving.

b) See 2.2, c).

c) A particle filter is an implementation of Bayesian filter and consists into assign to any position on the map a sample called particle. Each of these samples has a weight, that indicates the likelihood of the robot to be in that position. The weight then gets recursively updated while the robot is moving and senses more and more the environment. Its more important implementations are the Monte Carlo algorithm for localization and FastSLAM for SLAM.

d) Particles in the FastSLAM algorithm contain an estimated robots pose (x_t^k) and a set of Kalman Filters, with their mean and covariance, for each landmark that is on the map (we can say that there are N landmarks and M particles). At the time t , for each particle you'll have to take from the set described before its pose (x_{t-1}^k), predict the new pose at time t , update the measurements of the landmarks and then finally update the weight of the particle at the new time. After having done this "M" times, the final steps consists into resample with replacements all the M particles with a probability proportional to their weight.

4.2 Question:

Let's consider the group of gears in the picture where Gear A spins at 100 revolutions per minute driven by a 100W motor (recall $1W = 1N \times 1m / 1s$). (for the picture see on the pdf).

a) What are the output torque and speed on Gear D assuming no loss due to friction? b) Assume we are using a 24V motor with 0.8 efficiency, how much current is it driving?

Answer:

4.3 Question:

Describe the following kinematics and provide an example for each of them: a) Differential drive b) Synchronous drive c) Tricycle kinematics d) Ackerman steering

Answer:

a) A DD robot has the following composition: DRAW OF THE ROBOT. The wheel's velocities are the control variables, the robot can be driven only straight forward and so cannot translate laterally. *Put some formulas here... (Example: overboard?).*

b) A synchronous drive robot is composed by three wheels that are actuated and steered. DRAW OF THE ROBOT. There's a motor that makes them turn and one that makes them roll, and so it's possible to directly control θ . The wheels all point in the same direction and the control variables are the robot's linear and angular speed. The ICC is always at the infinite and the robot is holonomic. An example could be the Roomba.

c) Tricycle kinematics is composed like that: DRAW OF THE ROBOT. Its front wheel is the actuated one and it can steer with a maximum angle depending on the implementation, while the two posteriors are passive. It's the most diffused kinematics for Autonomous Guided Vehicles. The ICC is set on the line that passes through the passive wheels, control variables are the steering direction and the angular velocity of the front wheel. An example could be an industrial forklift.

d) Ackermann steering is the most diffused kinematics on the planet. DRAW OF THE ROBOT. It's composed by four wheels steering, with limited turning angles. In place rotation can't happen in this kinematics and the wheels have all their own velocity. It can be used the bicycle approximation for a two-wheel vehicle: in this case, the steering angle is unique and the formulas are simplified as there's no baseline between wheels. An example could be an autonomous driven car/motorbike.

4.4 Question:

What are nodes and messages in ROS and how are they used to develop a robot? Make an example of a simple robot and its possible ROS architecture.

Answer:

Nodes in ROS are processes that perform the computation. They are combined in a graph and communicate one another using topics and services. Messages are simple data structures that comprise typed fields. These are published on the topics by a single node and contain the information that it has to give to the others. A node that writes messages on a topic is called "Publisher", while one that reads from them is a "Subscriber" (it's better if it can accomplish both actions). *Example???*

4.5 Question:

Describe the Beam Sensor Model, its components, and the rationale behind each of them.

Answer:

See 1.4, a).

20/07/2016

5.1 Question:

Let's consider the Online SLAM problem, in particular: a) Provide the derivation of the recursive Bayes Filter for Online SLAM b) Describe the Bayes_filter algorithm for Online SLAM c) Describe the assumption in terms of uncertainty representation, motion model and measurement model behind the EKF-SLAM algorithm d) Describe the EKF-SLAM algorithm for Online SLAM

Answer:

a) See 2.2, e).

b)

c) Uncertainty Representation: EKF SLAM makes a Gaussian noise assumption for robot motion and perception. The amount of uncertainty in the posterior must be relatively small, since otherwise the linearization in EKFs tend to introduce intolerable errors.

Motion Model: maps in EKF SLAM are landmarks-based. For computational reasons, the number of landmarks is usually supposed small (< 1000) and further the EKF approach works well the less ambiguous the landmarks are.

Measurement Model: the EKF SLAM algorithm can only process positive sightings on landmarks, so it's supposed that there won't be negative measurements about the distance from them as they can't be processed.

d) The EKF SLAM algorithm starts by making a prediction about the mean and the covariance at time t associated to the robot's state (landmarks are included in this!), based on its knowledge of them at time $t-1$ and on the value of control variables. After that, for all the landmarks the robot has observed since now, if the landmark has never been seen before, its position gets initialised; it then calculates the Kalman Gain and so it uses it to correct the predictions made before. Finally, it updates the mean and the covariance using the corrected value and returns them.

5.2 Question:

Let's consider the problem of obstacle avoidance, a.k.a., local path planning, and the algorithms to implement it. Provide a description of a) What is the aim of local path planning b) The Vector Field Histogram approach (VFH) and its improvement Vector Field Histogram+ (VFH+) c) The Dynamic Window Approach (DWA) d) All these three algorithm use a navigation function, i.e., a function evaluating the value/cost of possible actions. Discuss briefly how it could be possible to set up the parameters involved in these functions

Answer:

See 1.1.

5.3 Question:

Describe the differences between messages, services, and actions in ROS.

Answer:

Messages are data structure which contain information that nodes in ROS want to exchange between one another; Services and Actions, instead, are not simply “boxes” with information as Messages are. Service and Actions are different between them because the first are similar to a call to a function and so are synchronous: when a service gets called by a client, the server has to send him a response: they have a smaller execution time than Actions and can’t be cancelled. Actions are more comparable to a thread paradigm and so have a longer execution time and, more importantly, are not synchronous: the node that made the request (client) can’t wait. They can also be cancelled.

5.4 Question:

What are direct and inverse kinematics of a robot? Provide an example of their use.

Answer:

See 2.1, c).

27/06/2016

6.1 Question:

Consider a differential robot base with a 50cm wheel base: a) Provide the direct kinematics formulas for a differential drive robot b) Assuming the robot starts in (0,0,0) compute the robot pose at times A, B and C according to the following actuations (numbers are strange, but computation should be easy) (picture in the pdf).

Answer:

Solutions on the pdf.

6.2 Question:

Let's consider the Simultaneous Localization and Mapping problem; answer the following questions a) How Simultaneous Localization and Mapping (SLAM) differs from Localization? b) How Simultaneous Localization and Mapping (SLAM) differs from Mapping? c) Describe what is the difference between online and offline SLAM d) Provide the derivation of the recursive Bayes Filter for online SLAM

Answer:

See 2.2.

6.3 Question:

Describe in detail the roles of the ROS Master in a ROS architecture (what it is, what it does).

Answer:

The ROS Master provides naming and registration services to the rest of the nodes in the ROS system. It tracks publishers and subscribers to topics as well as services, so it's essential for nodes interaction. The role of the Master is to enable individual ROS nodes to locate one another. Once these nodes have located each other they communicate with each other peer-to-peer. The Master also provides the Parameter Server. The Master is usually run using the "roscore" command, which loads the ROS Master along with other essential components, and so there's one Master for each system, even on distributed architectures. *(found on the wiki: if there's something more, notify)*